

# TCS NQT 2026

## Most repeated Number Based

### Coding Questions (Day-7)

#### 1 Program to Find the Sum of Digits of a Number

##### Problem Statement

Write a program to find the sum of digits of a given integer.

##### Example

Input: 1234

Output: 10

Explanation:  $1 + 2 + 3 + 4 = 10$

---

##### C++ Solution

```
#include <iostream>
using namespace std;
```

```
int main() {
    int n, sum = 0;
    cin >> n;

    while(n != 0) {
        sum += n % 10;
        n /= 10;
    }
}
```

```
    cout << sum;
    return 0;
}
```

### Java Solution

```
import java.util.Scanner;
```

```
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int sum = 0;

        while(n != 0) {
            sum += n % 10;
            n /= 10;
        }

        System.out.println(sum);
    }
}
```

### Python Solution

```
n = int(input())
sum = 0

while n != 0:
    sum += n % 10
    n //= 10
```

`print(sum)`

---

## **2 Program to Reverse a Number**

### **Problem Statement**

**Write a program to reverse a given integer.**

### **Example**

**Input: 1234**

**Output: 4321**

---

### **C++ Solution**

```
#include <iostream>
using namespace std;
```

```
int main() {
    int n, reverse = 0;
    cin >> n;

    while(n != 0) {
        reverse = reverse * 10 + n % 10;
        n /= 10;
    }

    cout << reverse;
    return 0;
}
```

### ☒ Java Solution

```
import java.util.Scanner;
```

```
public class Main {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
        int n = sc.nextInt();  
        int reverse = 0;  
  
        while(n != 0) {  
            reverse = reverse * 10 + n % 10;  
            n /= 10;  
        }  
  
        System.out.println(reverse);  
    }  
}
```

### ☒ Python Solution

```
n = int(input())  
reverse = 0  
  
while n != 0:  
    reverse = reverse * 10 + n % 10  
    n //= 10  
  
print(reverse)
```

---

### 3 Program to Check Whether a Number is Palindrome

#### Problem Statement

Write a program to check whether a number is a palindrome.

#### Example

Input: 121

Output: Palindrome

---

#### ☒ C++ Solution

```
#include <iostream>
using namespace std;
```

```
int main() {
    int n, original, reverse = 0;
    cin >> n;
    original = n;

    while(n != 0) {
        reverse = reverse * 10 + n % 10;
        n /= 10;
    }

    if(original == reverse)
        cout << "Palindrome";
    else
        cout << "Not Palindrome";
}
```

```
    return 0;  
}
```

### ☒ Java Solution

```
import java.util.Scanner;
```

```
public class Main {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
        int n = sc.nextInt();  
        int original = n;  
        int reverse = 0;  
  
        while(n != 0) {  
            reverse = reverse * 10 + n % 10;  
            n /= 10;  
        }  
  
        if(original == reverse)  
            System.out.println("Palindrome");  
        else  
            System.out.println("Not Palindrome");  
        }  
}
```

### ☒ Python Solution

```
n = int(input())  
original = n
```

```
reverse = 0
```

```
while n != 0:
```

```
    reverse = reverse * 10 + n % 10
```

```
    n //= 10
```

```
if original == reverse:
```

```
    print("Palindrome")
```

```
else:
```

```
    print("Not Palindrome")
```

---

#### **4 Program to Check Whether a Number is Prime**

**Problem Statement**

Write a program to check whether a given number is prime.

**Example**

**Input: 7**

**Output: Prime**

---

#### **☒ C++ Solution**

```
#include <iostream>
```

```
using namespace std;
```

```
int main() {
```

```
    int n;
```

```
    cin >> n;
```

```
if(n <= 1) {  
    cout << "Not Prime";  
    return 0;  
}  
  
for(int i = 2; i * i <= n; i++) {  
    if(n % i == 0) {  
        cout << "Not Prime";  
        return 0;  
    }  
}
```

```
cout << "Prime";  
return 0;  
}
```

☒ Java Solution

```
import java.util.Scanner;
```

```
public class Main {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
        int n = sc.nextInt();  
  
        if(n <= 1) {  
            System.out.println("Not Prime");  
            return;  
        }  
    }  
}
```



```

for(int i = 2; i * i <= n; i++) {
    if(n % i == 0) {
        System.out.println("Not Prime");
        return;
    }
}

System.out.println("Prime");
}
}

```

☒ Python Solution

```

n = int(input())

if n <= 1:
    print("Not Prime")
else:
    for i in range(2, int(n**0.5) + 1):
        if n % i == 0:
            print("Not Prime")
            break
    else:
        print("Prime")

```

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<https://youtube.com/playlist?list=PLAggmVp4R7cbiMDGsA62gnilXi0fOx0wz&si=-K-GJZdtziLKTkDL>

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## 5 Program to Find Sum of First N Prime Numbers

### Problem Statement

Write a program to find the sum of first n prime numbers.

### Example

Input: 3

Output: 10

Explanation:  $2 + 3 + 5 = 10$

---

### ☒ C++ Solution

```
#include <iostream>
using namespace std;
```

```
bool isPrime(int n) {
    if(n <= 1) return false;
    for(int i = 2; i * i <= n; i++)
        if(n % i == 0) return false;
    return true;
}
```

```
int main() {
    int n, count = 0, num = 2, sum = 0;
    cin >> n;
```

```
while(count < n) {  
    if(isPrime(num)) {  
        sum += num;  
        count++;  
    }  
    num++;  
}  
  
cout << sum;  
return 0;  
}
```

### Java Solution

```
import java.util.Scanner;  
  
public class Main {  
  
    static boolean isPrime(int n) {  
        if(n <= 1) return false;  
        for(int i = 2; i * i <= n; i++)  
            if(n % i == 0) return false;  
        return true;  
    }  
  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
        int n = sc.nextInt();  
        int count = 0, num = 2, sum = 0;
```

```
while(count < n) {  
    if(isPrime(num)) {  
        sum += num;  
        count++;  
    }  
    num++;  
}  
  
System.out.println(sum);  
}  
}
```

### Python Solution

```
def is_prime(n):  
    if n <= 1:  
        return False  
    for i in range(2, int(n**0.5) + 1):  
        if n % i == 0:  
            return False  
    return True
```

```
n = int(input())  
count = 0  
num = 2  
sum_primes = 0
```

```
while count < n:  
    if is_prime(num):  
        sum_primes += num
```

```
        count += 1
    num += 1

print(sum_primes)
```

---

## **6 Program to Find GCD and LCM of Two Numbers**

### **Problem Statement**

**Write a program to find GCD and LCM of two integers.**

### **Example**

**Input: 12 18**

**Output:**

**GCD = 6**

**LCM = 36**

---

### **C++ Solution**

```
#include <iostream>
using namespace std;
```

```
int main() {
    int a, b;
    cin >> a >> b;

    int x = a, y = b;

    while(b != 0) {
        int temp = b;
```

```
        b = a % b;
        a = temp;
    }

    int gcd = a;
    int lcm = (x * y) / gcd;

    cout << "GCD = " << gcd << endl;
    cout << "LCM = " << lcm;

    return 0;
}
```

## Java Solution

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int a = sc.nextInt();
        int b = sc.nextInt();

        int x = a, y = b;

        while(b != 0) {
            int temp = b;
            b = a % b;
            a = temp;
        }
    }
}
```

```
int gcd = a;
int lcm = (x * y) / gcd;

System.out.println("GCD = " + gcd);
System.out.println("LCM = " + lcm);
}
}
```

### Python Solution

```
a, b = map(int, input().split())
x, y = a, b
```

```
while b != 0:
    a, b = b, a % b
```

```
gcd = a
lcm = (x * y) // gcd
```

```
print("GCD =", gcd)
print("LCM =", lcm)
```

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## 7. Program to Check Armstrong Number

### Problem Statement

Write a program to check whether a given number is an Armstrong number.

An Armstrong number is a number that is equal to the sum of its digits raised to the power of the number of digits.

### Example

Input: 153

Output: Armstrong Number

Explanation:

$$1^3 + 5^3 + 3^3 = 1 + 125 + 27 = 153$$

---

### ☒ C++ Solution

```
#include <iostream>
```

```
#include <cmath>
```

```
using namespace std;
```

```
int main() {
```

```
    int n, original, remainder, result = 0, digits = 0;
```

```
    cin >> n;
```

```
    original = n;
```

```
    int temp = n;
```

```
    while(temp != 0) {
```

```
        digits++;
```



```
temp /= 10;  
}
```

```
temp = n;  
while(temp != 0) {  
    remainder = temp % 10;  
    result += pow(remainder, digits);  
    temp /= 10;  
}
```

```
if(result == original)  
    cout << "Armstrong Number";  
else  
    cout << "Not Armstrong Number";
```

```
return 0;  
}
```

☒ Java Solution

```
import java.util.Scanner;
```

```
public class Main {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
        int n = sc.nextInt();  
        int original = n, result = 0, digits = 0;  
  
        int temp = n;  
        while(temp != 0) {
```

```
        digits++;
        temp /= 10;
    }

    temp = n;
    while(temp != 0) {
        int remainder = temp % 10;
        result += Math.pow(remainder, digits);
        temp /= 10;
    }

    if(result == original)
        System.out.println("Armstrong Number");
    else
        System.out.println("Not Armstrong Number");
    }
}
```

#### ☒ Python Solution

```
n = int(input())
original = n
digits = len(str(n))
result = 0

while n != 0:
    remainder = n % 10
    result += remainder ** digits
    n //= 10
```

```
if result == original:
    print("Armstrong Number")
else:
    print("Not Armstrong Number")
```

---

## 8. Program to Find Sum of First N Natural Numbers

### Problem Statement

Write a program to find the sum of first n natural numbers.

Example

Input: 5

Output: 15

Explanation:  $1 + 2 + 3 + 4 + 5 = 15$

---

### ☒ C++ Solution

```
#include <iostream>
using namespace std;
```

```
int main() {
    int n;
    cin >> n;

    int sum = n * (n + 1) / 2;
    cout << sum;

    return 0;
}
```

### ☒ Java Solution

```
import java.util.Scanner;
```

```
public class Main {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
        int n = sc.nextInt();  
  
        int sum = n * (n + 1) / 2;  
        System.out.println(sum);  
    }  
}
```

### ☒ Python Solution

```
n = int(input())  
sum = n * (n + 1) // 2  
print(sum)
```

---

## 9. Program to Find Factorial of a Number

### Problem Statement

Write a program to find factorial of a number.

Factorial of  $n = n \times (n-1) \times (n-2) \times \dots \times 1$

### Example

Input: 5

Output: 120

Explanation:  $5 \times 4 \times 3 \times 2 \times 1 = 120$

---

☒ C++ Solution

```
#include <iostream>
using namespace std;
```

```
int main() {
    int n;
    cin >> n;

    long long fact = 1;
    for(int i = 1; i <= n; i++) {
        fact *= i;
    }

    cout << fact;
    return 0;
}
```

☒ Java Solution

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();

        long fact = 1;
        for(int i = 1; i <= n; i++) {
```

```
        fact *= i;
    }

    System.out.println(fact);
}
}
```

### ✓ Python Solution

```
n = int(input())
fact = 1
```

```
for i in range(1, n + 1):
    fact *= i
```

```
print(fact)
```

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---

## 10. Program to Check Perfect Number

### Problem Statement

Write a program to check whether a number is a Perfect Number.

A number is perfect if the sum of its proper divisors equals the number.

**Example**

**Input: 6**

**Output: Perfect Number**

**Explanation:**

**Divisors: 1, 2, 3**

**Sum = 1 + 2 + 3 = 6**

---

☒ **C++ Solution**

```
#include <iostream>
using namespace std;
```

```
int main() {
    int n;
    cin >> n;

    int sum = 0;
    for(int i = 1; i <= n/2; i++) {
        if(n % i == 0)
            sum += i;
    }

    if(sum == n)
        cout << "Perfect Number";
    else
        cout << "Not Perfect Number";
}
```

```
    return 0;
}
```

### ☒ Java Solution

```
import java.util.Scanner;
```

```
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();

        int sum = 0;
        for(int i = 1; i <= n/2; i++) {
            if(n % i == 0)
                sum += i;
        }

        if(sum == n)
            System.out.println("Perfect Number");
        else
            System.out.println("Not Perfect Number");
        }
}
```

### ☒ Python Solution

```
n = int(input())
sum_div = 0
```



```
for i in range(1, n // 2 + 1):
    if n % i == 0:
        sum_div += i

if sum_div == n:
    print("Perfect Number")
else:
    print("Not Perfect Number")
```

---

## 11. Program to Check Prime Number

### Problem Statement

Write a program to check whether a number is Prime.

Example

Input: 7

Output: Prime Number

---

### ☒ C++ Solution

```
#include <iostream>
using namespace std;
```

```
int main() {
    int n;
    cin >> n;

    if(n <= 1) {
        cout << "Not Prime";
```

```
    return 0;
}

for(int i = 2; i * i <= n; i++) {
    if(n % i == 0) {
        cout << "Not Prime";
        return 0;
    }
}
```

```
    cout << "Prime Number";
    return 0;
}
```

☒ Java Solution

```
import java.util.Scanner;
```

```
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();

        if(n <= 1) {
            System.out.println("Not Prime");
            return;
        }
    }
}
```

```
    for(int i = 2; i * i <= n; i++) {
        if(n % i == 0) {
```

```
        System.out.println("Not Prime");
        return;
    }
}
```

```
        System.out.println("Prime Number");
    }
}
```

☒ Python Solution

```
n = int(input())
```

```
if n <= 1:
```

```
    print("Not Prime")
```

```
else:
```

```
    for i in range(2, int(n**0.5) + 1):
```

```
        if n % i == 0:
```

```
            print("Not Prime")
```

```
            break
```

```
    else:
```

```
        print("Prime Number")
```